Heba Qasem

Computer Systems Engineering

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EDUCATION

B.Sc of Computer Systems Engineering 2021 – 06/2025
Palestine Technical University - Kadoorie (PTUK). Tulkarm
Average 86.7%.

Tawjihi (Scientific Stream) 2020 – 2021
Jalqamous Secondary School. Jenin
Average 98.4%.

PROFESSIONAL EXPERIENCE

AI & Software Engineering Experience (Academic & Projects)

- Implemented machine learning models using **XGBoost, Random Forest, and PyCaret** to predict continuous values from structured datasets.
- Performed **data preprocessing, feature encoding, and model evaluation** using Python, Pandas, Scikit-learn, and Matplotlib.
- Strengthened knowledge of **data structures and algorithms** by solving **145+ problems on Codeforces** and **75+ problems on LeetCode**.

PROJECTS

PPV Value Prediction using XGBoost
Python, XGBoost, Scikit-learn, Pandas

- Built a regression model to predict PPV values from demographic and duration data (5,270 records).
- Applied preprocessing, categorical encoding, and feature importance analysis.
- Evaluated performance using **MAE, RMSE, and R^2**.

PPV Value Prediction using PyCaret
Python, PyCaret, Random Forest, Scikit-learn

- Developed an **automated ML pipeline** using PyCaret to compare and train multiple regression models.
- Selected **Random Forest Regressor** as best-performing model using compare_models().

SKILLS

Programming & Software Engineering

- Python, C#, Java, SQL, JavaScript
- Object-Oriented Programming (OOP), Data Structures, Algorithms, Design Patterns

Machine Learning & AI

- Supervised Learning (Regression, Classification), Model Evaluation (MAE, RMSE, R^2)
- Libraries: Scikit-learn, XGBoost, PyCaret, Pandas, NumPy
- Deep Learning Basics: TensorFlow, Keras

Tools & Platforms

- Git/GitHub, Jupyter Notebook, Visual Studio, VS Code
- SQL Server, MySQL

LANGUAGES

English (Professional) Arabic (Native)